

**MUSOMA, Tanzania, United Republic of**WMO: **637330**Lat: **1.4962S**Lon: **33.8023E**Elev: **1147**StdP: **88.28**Time Zone: **3.00 (E03)**Period: **06-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | 17.1       | 17.9 | 10.5                          | 9.1 | 23.5 | 11.7 | 9.8 | 23.7 | 8.3                   | 25.5 | 7.5 | 26.4 | 0.2                   | 170  | 0.244 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 8.7                    | 30.8            | 19.8 | 30.0 | 20.1 | 29.2 | 20.3 | 22.6                | 27.5 | 22.2 | 27.1 | 22.0 | 26.8 | 3.9                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 21.2                            | 18.2 | 24.9 | 20.9 | 17.9 | 24.7 | 20.4 | 17.4 | 24.3 | 73.2          | 27.6 | 71.4 | 27.2 | 70.3 | 26.9 | 27.0              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |  |  |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|--|--|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |  |  |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |  |  |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |  |  |
| (4) 7.8           | 6.8  | 6.1 | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
| (5)               |      |     | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 23.7   | 23.8  | 24.4  | 24.3  | 23.9  | 23.6  | 23.2  | 23.0  | 23.5  | 23.9  | 24.2  | 23.5  | 23.2 |
|                                                                                  | DBStd       | 1.15   | 1.07  | 1.22  | 1.34  | 1.08  | 0.85  | 0.90  | 0.89  | 1.03  | 1.10  | 1.12  | 1.02  | 1.09 |
|                                                                                  | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | HDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDD10.0     | 5002   | 429   | 404   | 443   | 416   | 423   | 397   | 402   | 417   | 418   | 439   | 405   | 408  |
|                                                                                  | CDD18.3     | 1961   | 171   | 171   | 185   | 166   | 164   | 147   | 144   | 159   | 168   | 181   | 155   | 150  |
|                                                                                  | CDH23.3     | 11010  | 1008  | 1142  | 1198  | 923   | 877   | 794   | 811   | 858   | 933   | 1018  | 757   | 690  |
| CDH26.7                                                                          | 1972        | 189    | 281   | 301   | 178   | 134   | 109   | 104   | 130   | 162   | 198   | 107   | 79    |      |
| Wind                                                                             | WSAvg       | 2.1    | 1.9   | 2.1   | 2.2   | 1.8   | 1.7   | 1.8   | 2.1   | 2.3   | 2.5   | 2.3   | 2.0   | 1.8  |
| Precipitation                                                                    | PrecAvg     | 889    | 60    | 80    | 117   | 188   | 108   | 26    | 16    | 15    | 30    | 62    | 108   | 78   |
|                                                                                  | PrecMax     | 1430   | 179   | 250   | 399   | 323   | 224   | 98    | 98    | 53    | 121   | 221   | 500   | 269  |
|                                                                                  | PrecMin     | 507    | 1     | 8     | 19    | 73    | 31    | 0     | 0     | 0     | 2     | 10    | 17    | 7    |
|                                                                                  | PrecStd     | 193    | 44    | 60    | 81    | 70    | 51    | 25    | 21    | 14    | 28    | 46    | 86    | 58   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 30.8  | 31.3  | 31.8  | 30.8  | 30.0  | 29.5  | 29.5  | 30.2  | 30.2  | 31.1  | 30.1  | 29.7 |
|                                                                                  |             | MCWB   | 19.7  | 19.7  | 19.8  | 20.6  | 20.5  | 19.6  | 18.6  | 18.7  | 19.4  | 19.0  | 19.9  | 20.0 |
|                                                                                  | 2%          | DB     | 29.3  | 30.1  | 30.1  | 29.2  | 29.0  | 28.8  | 28.4  | 29.0  | 29.1  | 29.7  | 28.8  | 28.2 |
|                                                                                  |             | MCWB   | 20.2  | 20.2  | 20.6  | 21.1  | 20.7  | 19.7  | 19.2  | 19.0  | 19.6  | 19.7  | 20.4  | 20.4 |
|                                                                                  | 5%          | DB     | 28.7  | 29.2  | 29.2  | 28.5  | 28.1  | 28.0  | 27.9  | 28.0  | 28.2  | 28.8  | 27.9  | 27.7 |
|                                                                                  |             | MCWB   | 20.5  | 20.5  | 20.8  | 21.2  | 20.9  | 20.2  | 19.4  | 19.6  | 19.9  | 20.1  | 20.6  | 20.5 |
|                                                                                  | 10%         | DB     | 27.8  | 28.4  | 28.4  | 27.8  | 27.2  | 27.1  | 27.1  | 27.1  | 27.5  | 27.8  | 27.1  | 26.9 |
|                                                                                  |             | MCWB   | 20.7  | 20.7  | 21.0  | 21.1  | 20.8  | 20.2  | 19.5  | 19.7  | 20.1  | 20.4  | 20.7  | 20.7 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 22.7  | 23.0  | 23.0  | 22.9  | 22.7  | 22.1  | 21.7  | 22.1  | 22.1  | 22.4  | 22.5  | 22.3 |
|                                                                                  |             | MCDB   | 27.4  | 27.7  | 28.3  | 27.8  | 27.6  | 27.0  | 26.5  | 27.0  | 27.2  | 27.1  | 26.8  | 26.8 |
|                                                                                  | 2%          | WB     | 22.1  | 22.3  | 22.3  | 22.2  | 22.0  | 21.6  | 21.1  | 21.2  | 21.6  | 21.9  | 21.9  | 21.8 |
|                                                                                  |             | MCDB   | 26.9  | 27.2  | 27.5  | 27.2  | 27.1  | 26.4  | 26.1  | 26.0  | 26.5  | 26.5  | 26.3  | 26.2 |
|                                                                                  | 5%          | WB     | 21.7  | 21.9  | 21.9  | 21.9  | 21.6  | 21.2  | 20.7  | 20.9  | 21.2  | 21.5  | 21.5  | 21.4 |
|                                                                                  |             | MCDB   | 26.3  | 26.9  | 27.0  | 26.8  | 26.4  | 25.8  | 25.6  | 25.7  | 26.0  | 26.1  | 25.8  | 25.6 |
|                                                                                  | 10%         | WB     | 21.2  | 21.5  | 21.5  | 21.5  | 21.2  | 20.7  | 20.3  | 20.6  | 20.9  | 21.1  | 21.1  | 21.1 |
|                                                                                  |             | MCDB   | 25.8  | 26.4  | 26.5  | 26.2  | 25.8  | 25.3  | 25.1  | 25.2  | 25.6  | 25.6  | 25.4  | 25.2 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.3   | 8.7   | 8.7   | 8.4   | 8.6   | 8.8   | 9.5   | 9.1   | 8.5   | 8.5   | 8.0   | 7.4  |
|                                                                                  |             | MCDBR  | 9.6   | 9.8   | 10.0  | 9.7   | 9.5   | 9.9   | 10.4  | 10.1  | 9.4   | 9.9   | 9.2   | 8.7  |
|                                                                                  |             | MCWBR  | 4.0   | 4.4   | 4.3   | 4.0   | 4.0   | 4.6   | 4.9   | 5.0   | 4.6   | 4.1   | 3.8   | 3.6  |
|                                                                                  | 5% WB       | MCDBR  | 8.5   | 8.7   | 9.2   | 8.6   | 8.9   | 8.8   | 9.4   | 9.1   | 8.6   | 8.7   | 8.3   | 7.8  |
|                                                                                  |             | MCWBR  | 3.9   | 4.2   | 4.2   | 3.9   | 4.0   | 4.3   | 4.6   | 4.9   | 4.4   | 4.0   | 3.7   | 3.5  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.419  | 0.435 | 0.422 | 0.370 | 0.366 | 0.418 | 0.437 | 0.428 | 0.396 | 0.389 | 0.384 | 0.410 |      |
|                                                                                  | taud        | 2.367  | 2.314 | 2.388 | 2.555 | 2.555 | 2.358 | 2.264 | 2.304 | 2.402 | 2.426 | 2.476 | 2.397 |      |
|                                                                                  | Ebn at Noon | 913    | 901   | 904   | 930   | 910   | 847   | 836   | 864   | 913   | 931   | 940   | 917   |      |
|                                                                                  | Edh at Noon | 129    | 137   | 127   | 104   | 100   | 119   | 132   | 132   | 123   | 121   | 114   | 124   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 5.88   | 6.31  | 6.26  | 5.89  | 5.76  | 5.51  | 5.56  | 5.68  | 6.00  | 6.01  | 5.68  | 5.67  |      |
|                                                                                  | RadStd      | 0.49   | 0.33  | 0.45  | 0.39  | 0.26  | 0.35  | 0.33  | 0.29  | 0.31  | 0.30  | 0.31  | 0.39  |      |

**Historical Trends**

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (2 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page